Online Appendix

Multiple membership configurations and trust formation:
Structural precarity in highly unequal societies

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S1 Attrition Analysis and Sample Comparisons

We assess whether the balanced panel ($n = 1{,}304$ respondents observed in all three waves) differs systematically from the 1,623 respondents who participated in Wave 1 but were subsequently lost to attrition. Table A1 compares Wave 1 (2016) baseline characteristics.

Table A1: Baseline characteristics: balanced panel vs. attrited respondents (Wave 1, 2016)

	Balanced panel		Attrited		
	Mean	SD	Mean	SD	p
Age (years)	47.2	14.6	45.2	15.8	$***$
Woman (%)	66.3	—	55.5	—	$***$
Secondary education or higher (%)	64.3	—	63.8	—	
Employed (%)	57.0	—	61.6	—	$*$
Generalized trust (%)	7.7	—	10.8	—	$**$
Neighbourhood trust (%)	48.5	—	44.9	—	†
Membership domains (mean)	1.25	1.35	1.08	1.25	$***$
N	1,304		1,623		

Notes. Wave 1 (2016) baseline comparisons. Balanced panel: observed in all three waves (2016, 2018, 2022). Attrited: present in Wave 1 but missing in at least one subsequent wave. p-values: two-sample t-tests (continuous) or proportion tests (binary). $\dagger p < .10$; $*p < .05$; $**p < .01$; $***p < .001$.

The retained sample is on average slightly older (+2.0 years), more female (+10.8 pp), reports lower generalized trust (−3.1 pp), and has higher associational breadth (+0.17 domains). The education and employment differences are small (< 1 pp and −4.6 pp, respectively). These differences suggest selective retention of respondents who are more socially embedded and somewhat less trusting. We note that the direction of the trust difference is conservative with respect to our main hypotheses: if bridgers are more trusting and also more likely to be retained, any bias would *inflate* the estimated position–trust association. Our reported estimates are therefore likely conservative lower bounds.

S2 Latent Markov Model: Diagnostics and Robustness

S2.1 Model Fit Statistics

Table A2 reports fit statistics for $K = 1, \ldots, 5$ latent states. The BIC selects $K = 3$, with a $\Delta\text{BIC} = -632$ relative to $K = 2$. The $K = 4$ solution yields $\Delta\text{BIC} = +183$, indicating that a fourth state adds complexity without improving fit.

Table A2: Latent Markov model fit statistics ($K = 1$ to 5)

K	Log-likelihood	Parameters	AIC	BIC	ΔBIC
1	−11,757.84	8	23,531.67	23,573.01	—
2	−11,113.77	46	22,319.55	22,557.27	−1,015.75
3	−10,589.93	104	21,387.87	21,925.32	−631.95
4	−10,401.95	182	21,167.89	22,108.43	+183.11
5	−10,208.13	280	20,976.27	22,423.26	+314.82

Notes. Bold row: selected model ($K = 3$, minimum BIC). ΔBIC relative to the previous row. Estimated via EM algorithm (`LMest` package, R). ELSOC balanced panel (2016, 2018, 2022); $n = 1{,}304$ individuals, three waves.

S2.2 Classification Quality

Table A3 reports classification diagnostics for the $K = 3$ solution. The mean maximum posterior probability is 0.885, and 92.7% of observations are strictly classified ($p_{\max} \geq .60$), indicating high separation between latent states. Normalized entropy is 0.264 (lower = better separation; minimum is 0).

Table A3: Classification quality by assigned position ($K = 3$)

Assigned position	n	% of obs.	Avg. $\hat{P}$(assigned state)
α (Isolate)	2,160	55.5%	0.894
β (Closed)	1,334	34.3%	0.876
γ (Bridging)	397	10.2%	0.868
Overall mean $p_{\max}$		0.885	
Share strictly classified ($p_{\max} \geq .60$)		92.7%	
Normalized entropy		0.264	

Notes. Average posterior probability of the assigned state for each modally-assigned position (diagonal of the posterior probability matrix). Values > 0.70 are generally considered satisfactory. Total: 3,891 person-wave observations; $n_{\text{id}} = 1{,}297$ unique respondents.

S2.3 Latent Portfolio Profiles

Table A4 reports the full item-response probability matrix from the $K = 3$ model. α (Isolate) shows uniformly low membership probability (< 0.20) across all eight domains. β (Closed) shows elevated probability in neighborhood and religious organizations but low probability elsewhere (2 domains $\geq .30$; diversity $D = 0.25$). γ (Bridging) shows high membership probability across six of eight domains (diversity $D = 0.75$).

Figure A1 displays these profiles graphically.

Table A4: Item-response probabilities: Pr(member | state) by domain and position

Domain	α (Isolate)	β (Closed)	γ (Bridging)
Neighborhood organizations (JJVV)	0.18	0.60	0.61
Religious organizations	0.08	0.67	0.48
Political parties	0.02	0.03	0.28
Labor unions	0.09	0.03	0.39
Professional associations	0.03	0.01	0.45
Charitable organizations	0.05	0.13	0.54
Sports clubs	0.11	0.05	0.40
Student organizations	0.02	0.02	0.24
Domains with Pr $\geq .30$	0	2	6
Domain diversity ($D = C/8$)	0.00	0.25	0.75

Notes. State-specific conditional response probabilities from the $K = 3$ Latent Markov Model. State mapping: α = State 1, γ = State 2, β = State 3. Domain diversity $D = C/C_{\max}$, where C = domains with Pr $\geq .30$ and $C_{\max} = 8$.

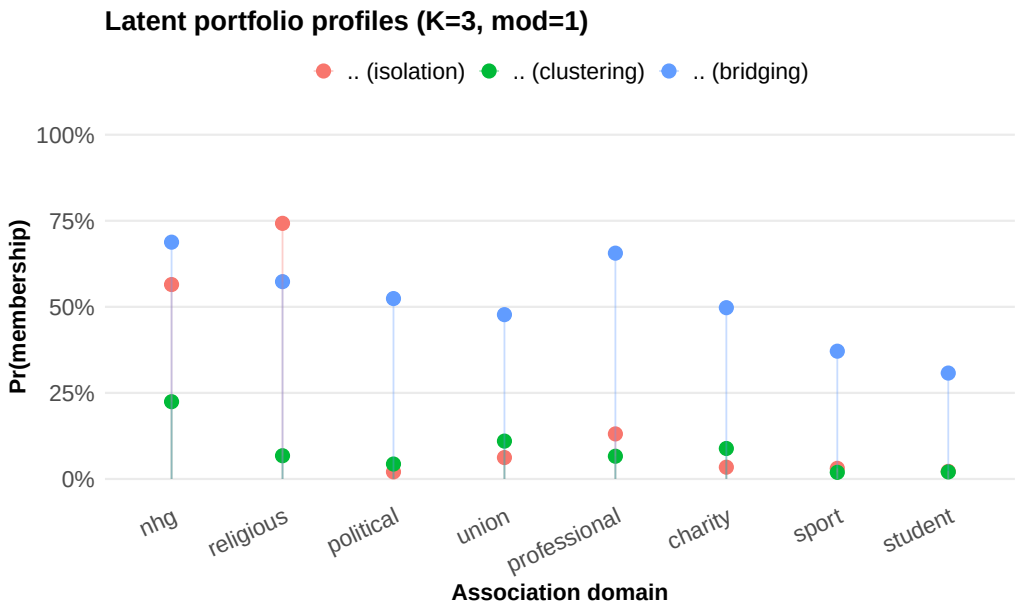

Figure A1: Latent portfolio profiles ($K = 3$ Latent Markov Model). Each bar shows the probability of membership in the indicated organizational domain. α (Isolate): uniformly low membership. β (Closed): concentrated in neighbourhood and religious organizations. γ (Bridging): high membership probability across multiple domains.

S3 H3: Portfolio Persistence and Structural Precarity

S3.1 Persistence by position and transition period

Table A5 decomposes portfolio-position persistence by starting position and transition period. The exit probability from γ is approximately 40% in both wave-transitions, compared with under 12% for α and β, confirming that the instability of the bridging position is not a Wave-specific artefact.

Table A5: Portfolio-position persistence by transition period

Position	Transition	n	P(stay)	P(exit)	95% CI
α (Isolate)	2016→2018	720	0.911	0.089	[0.890, 0.932]
α (Isolate)	2018→2022	717	0.933	0.067	[0.915, 0.951]
β (Closed)	2016→2018	448	0.886	0.114	[0.857, 0.916]
β (Closed)	2018→2022	436	0.929	0.071	[0.905, 0.953]
γ (Bridging)	2016→2018	129	0.597	0.403	[0.512, 0.682]
γ (Bridging)	2018→2022	144	0.576	0.424	[0.496, 0.657]

Notes. Unconditional persistence and exit rates by position and transition period. 95% CI based on normal approximation. n: person-wave observations per cell.

S3.2 SES predictors of exit from γ: four specifications

Table A6: Socioeconomic predictors of exit from γ: probit AMEs (four specifications)

	Outcome: P(exit from γ at $t+1$)			
	M1: Edu only	M2: Emp only	M3: Full	M4: Edu ordinal
High education (ref: low)	-0.229^{***}		-0.220^{**}	
	(0.065)		(0.065)	
Education (ordinal, 1–4)				-0.113^{***}
				(0.025)
Employed (ref: other)		$-0.130^{\dagger}$	-0.149^{*}	-0.148^{*}
		(0.067)	(0.067)	(0.069)
n			273	

Notes. AMEs from probit models. Sample: person-wave observations in γ at t. M1–M3: binary high education (técnica or university vs. lower); M4: ordinal education (1–4). All models include sex, age, wave indicator. Cluster-robust SE (by individual) in parentheses. $\dagger p < .10$; $*p < .05$; $**p < .01$; $***p < .001$.

S3.3 Specificity test: exit models by starting position

Table A7: Socioeconomic predictors of exit by starting position (specificity test)

Predictor	AME	(SE)	p-value
Starting position: α (Isolate) — exit rate = 7.8%, $n = 1{,}437$			
High education	$+0.029^{\dagger}$	(0.016)	0.065
Employed	$+0.013$	(0.015)	0.364
Starting position: β (Closed) — exit rate = 9.3%, $n = 884$			
High education	$+0.082^{*}$	(0.035)	0.019
Employed	-0.008	(0.020)	0.682
Starting position: γ (Bridging) — exit rate = 41.4%, $n = 273$			
High education	-0.220^{***}	(0.065)	0.001
Employed	-0.149^{*}	(0.067)	0.026

Notes. AMEs from probit models of P(exit from starting position at $t+1$). Each panel uses respondents starting in the indicated position. Controls: sex, age, wave indicator. Cluster-robust SE (by individual). $\dagger p < .10$; $*p < .05$; $**p < .01$; $***p < .001$.

S3.4 Cluster bootstrap validation

Given the modest size of the γ subsample ($n_{\text{id}} = 196$ unique respondents), we validate the AMEs using a cluster bootstrap ($B = 999$) that resamples individuals with replacement. Table A8 shows that bootstrap standard errors are virtually identical to cluster-robust analytical SEs (ratio ≈ 1.0), and zero lies outside both confidence intervals.

Table A8: Cluster bootstrap validation of AMEs for exit from γ

		SE		95% CI	
Predictor	AME	Cluster-robust	Bootstrap	Cluster-robust	Bootstrap
High education (ref: low)	-0.220	0.065	0.065	$[-0.348, -0.093]$	$[-0.346, -0.087]$
Employed (ref: other)	-0.149	0.067	0.068	$[-0.281, -0.018]$	$[-0.284, -0.014]$

Notes. $n = 273$ person-wave observations; $n_{\text{id}} = 196$ unique respondents. Bootstrap: $B = 999$ cluster resamples (ids sampled with replacement). CI: percentile method [2.5%, 97.5%]. Cluster-robust SE: `vcovCL` sandwich.

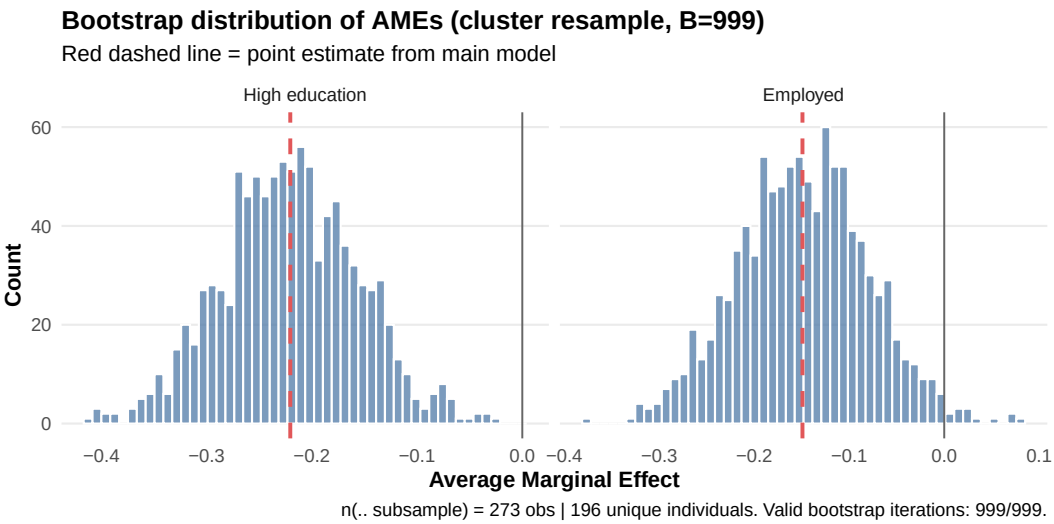

Figure A2: Bootstrap distributions of AMEs for exit from γ ($B = 999$ cluster resamples). Red dashed line: point estimate from M3. Both distributions are approximately normal; zero lies outside both.

S4 Trust Models: Main Specification

Table A9 reports average marginal effects (AMEs) from the random-effects probit models that constitute our preferred specification (Models 1–4 of Table 4 in the main text).

Table A9: Random-effects probit models: AMEs on trust (full results)

	Generalized trust		Neighbourhood trust	
	M1 (no ctrl)	M2 (ctrl)	M3 (no ctrl)	M4 (ctrl)
β vs. γ	-0.059^{**}	-0.051^{**}	-0.037	-0.069^{*}
	(0.018)	(0.018)	(0.035)	(0.033)
α vs. γ	-0.035^{*}	$-0.031^{\dagger}$	-0.091^{**}	-0.090^{**}
	(0.018)	(0.017)	(0.033)	(0.030)
N (person-waves)		3,891		
n (individuals)		1,297		
Wave FE	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes

Notes. AMEs from RE probit (GHQ-12 quadrature, nAGQ = 12). Reference category: γ (Bridging). Controls: age, sex, subjective wellbeing, employment, couple status, institutional trust. SE in parentheses (model-based with cluster-robust fallback). $\dagger p < .10$; $*p < .05$; $**p < .01$; $***p < .001$.

S4.1 Reverse causation test

To assess whether trust predicts subsequent portfolio position (rather than the reverse), Table A10 reports linear random-effects models with portfolio position (numeric) as outcome and trust as predictor. None of the four estimates approaches conventional significance thresholds, providing no evidence of reverse causation.

Table A10: Reverse relationship: trust predicting portfolio position

Model	AME	(SE)	p-value
Generalized trust, no controls	+0.020	(0.029)	.486
Generalized trust, with controls	+0.019	(0.029)	.510
Neighbourhood trust, no controls	−0.028	(0.017)	.101
Neighbourhood trust, with controls	−0.018	(0.017)	.313

Notes. AMEs from linear random-effects models of portfolio position (numeric: $1 = \alpha$, $2 = \beta$, $3 = \gamma$) predicted by trust. Wave fixed effects included in all models. N = 3,891 person-waves; n = 1,297 individuals. $\dagger p < .10$; $*p < .05$; $**p < .01$; $***p < .001$.

S5 Sensitivity Analyses for Trust Models

We report five pre-planned sensitivity analyses. Table A11 summarizes all significant results; Sections S5.1–S5.5 provide full detail. The central finding — β vs. γ for generalized trust — is significant in all four model-based specifications (S1–S4) with no sign reversals.

Table A11: Sensitivity analyses: summary of significant AMEs ($p < .10$)

Specification	Outcome	Contrast	AME (SE)
S1: CRE / Mundlak pooled probit			
	Generalized trust	β vs. γ	-0.077^{***} (0.023)
	Generalized trust	α vs. γ	-0.054^{*} (0.022)
	Neighbourhood trust	β vs. γ	-0.072^{*} (0.033)
	Neighbourhood trust	α vs. γ	-0.088^{**} (0.030)
S3: Cross-lagged			
	Generalized trust	β vs. γ	-0.066^{**} (0.022)
	Generalized trust	α vs. γ	-0.048^{*} (0.021)
	Neighbourhood trust	β vs. γ	-0.064^{*} (0.032)
	Neighbourhood trust	α vs. γ	-0.063^{*} (0.029)
S4: Entropy-weighted pooled probit			
	Generalized trust	β vs. γ	-0.090^{***} (0.024)
	Generalized trust	α vs. γ	-0.060^{*} (0.024)
	Neighbourhood trust	β vs. γ	-0.081^{*} (0.033)
	Neighbourhood trust	α vs. γ	-0.096^{**} (0.030)
S5: Soft assignment (posterior probabilities)			
	Generalized trust	p_β (1-unit)	-0.041^{**} (0.013)
	Generalized trust	p_γ (1-unit)	$+0.029^{*}$ (0.014)
	Neighbourhood trust	p_γ (1-unit)	$+0.119^{***}$ (0.036)

Notes. Reference: γ (Bridging) in S1–S4. In S5, p_γ and p_β are continuous predictors (implicit reference: p_α). Main model (Table A9): β vs. γ generalized trust AME $= -0.051^{**}$. S2 (Oster δ) produces bounds, not AMEs; see Table A13. $\dagger p < .10$; $*p < .05$; $**p < .01$; $***p < .001$.

S5.1 S1: Correlated Random Effects / Mundlak Probit

Standard RE probit assumes that unobserved individual heterogeneity is uncorrelated with covariates. We relax this using the Mundlak (1978) / Wooldridge (2019) correction: a pooled probit augmented with within-person means of time-varying covariates, estimated with cluster-robust SEs (`vcovCL` by respondent). This yields population-average partial effects (APEs) without the conservative shrinkage of RE-GHQ models.

A Wald test of the joint significance of the mean terms rejects H_0 for both outcomes (generalized trust: $\chi^2(4) = 16.43$, $p = .002$; neighbourhood trust: $\chi^2(4) = 40.91$, $p < .001$), confirming detectable endogeneity bias in the standard RE model. Under CRE correction, effects grow in magnitude, indicating that main estimates are conservative.

Table A12: S1: Correlated Random Effects (Mundlak) pooled probit

Outcome	Contrast	AME	SE	p
Generalized trust	β vs. γ	-0.077	0.023	.001***
Generalized trust	α vs. γ	-0.054	0.022	.015*
Neighbourhood trust	β vs. γ	-0.072	0.033	.027*
Neighbourhood trust	α vs. γ	-0.088	0.030	.003**

Notes. Pooled probit with within-person means of *swb*, *employed*, *couple*, *tinst* as Mundlak corrections. Cluster-robust SE by respondent (`vcovCL`). Wald test of mean terms: $\chi^2(4) = 16.43$ ($p = .002$, generalized trust); $\chi^2(4) = 40.91$ ($p < .001$, neighbourhood trust). Reference: γ (Bridging). $N = 3{,}891$. $\dagger p < .10$; $*p < .05$; $**p < .01$; $***p < .001$.

S5.2 S2: Oster (2019) δ Bounds

The Oster (2019) test asks how strongly an omitted variable would need to correlate with both treatment and outcome to fully explain the estimated effect. Setting $R^2_{\max} = \min(1.3 \times R^2_{\text{full}}, 1)$, all four contrasts yield $|\delta| > 1$ (Table A13), satisfying the conventional robustness threshold.

Table A13: S2: Oster (2019) δ bounds

Outcome	Contrast	$\hat{\beta}_r$	$\hat{\beta}_f$	R^2_r	R^2_f	$R^2_{\max}$	δ
Gen. trust	β vs. γ	-0.099	-0.087	0.013	0.019	0.025	6.72
Gen. trust	α vs. γ	-0.067	-0.063	0.013	0.019	0.025	16.22
NH trust	β vs. γ	-0.060	-0.078	0.006	0.104	0.135	-1.35
NH trust	α vs. γ	-0.109	-0.095	0.006	0.104	0.135	2.14

Notes. OLS approximation on pooled data. $\hat{\beta}_r$: without controls; $\hat{\beta}_f$: with full controls. $R^2_{\max} = \min(1.3R^2_f, 1.0)$. $\delta < -1$ for neighbourhood trust β vs. γ indicates that adding controls *amplified* the effect (conservative direction). (Oster, 2019).

S5.3 S3: Cross-Lagged Specification

Position at wave $t-1$ predicts trust at wave t, controlling for lagged trust at $t-1$. Restricted to waves 2 and 3 ($N = 2{,}594$; $n_{\text{id}} = 1{,}297$). All four contrasts remain significant (Table A14). Singular-fit warnings reflect expected collapse of RE variance when a lagged DV is included; AMEs remain valid.

Table A14: S3: Cross-lagged AMEs — position at $t-1$ predicts trust at t, controlling for trust at $t-1$

Outcome	Contrast	AME	SE	p
Generalized trust	β vs. γ	-0.066	0.022	$.002^{**}$
Generalized trust	α vs. γ	-0.048	0.021	$.019^{*}$
Neighbourhood trust	β vs. γ	-0.064	0.032	$.045^{*}$
Neighbourhood trust	α vs. γ	-0.063	0.030	$.034^{*}$

Notes. RE probit (GHQ-12, nAGQ = 12). Waves 2–3 only (N = 2,594; n_{id} = 1,297). Lagged trust at $t-1$ included as control. Reference: γ (Bridging). $\dagger p < .10$; $*p < .05$; $**p < .01$; $***p < .001$.

S5.4 S4: Entropy-Weighted Pooled Probit

Observations are weighted by $w_i = p_{\max,i}/\bar{p}_{\max}$ (mean = 1), downweighting uncertain classifications. We use a pooled probit with frequency weights and cluster-robust SEs. Mean $p_{\max} = 0.885$. All contrasts remain significant and grow in magnitude relative to the main model (Table A15), consistent with attenuation bias from classification noise in the main specification.

Table A15: S4: Entropy-weighted pooled probit AMEs

Outcome	Contrast	AME	SE	p
Generalized trust	β vs. γ	-0.090	0.024	$< .001^{***}$
Generalized trust	α vs. γ	-0.060	0.024	$.011^{*}$
Neighbourhood trust	β vs. γ	-0.081	0.033	$.015^{*}$
Neighbourhood trust	α vs. γ	-0.096	0.030	$.002^{**}$

Notes. Pooled probit with importance weights $w_i = p_{\max,i}/\bar{p}_{\max}$. Cluster-robust SE by respondent (`vcovCL`). N = 3,891. Reference: γ (Bridging). $\dagger p < .10$; $*p < .05$; $**p < .01$; $***p < .001$.

S5.5 S5: Soft Assignment with Posterior Probabilities

Posterior probabilities p_γ and p_β replace modal class assignment as continuous predictors (reference: p_α, omitted). Higher p_γ predicts more trust in both outcomes; higher p_β predicts less generalized trust (Table A16). The neighbourhood trust effect of p_γ ($+0.119^{***}$, SE = 0.036) is the largest point estimate in the sensitivity analysis.

Table A16: S5: Soft-assignment AMEs — posterior probabilities as continuous predictors

Outcome	Predictor	AME	SE	p
Generalized trust	p_β (1-unit increase)	-0.041	0.013	$.002^{**}$
Generalized trust	p_γ (1-unit increase)	$+0.029$	0.014	$.037^{*}$
Neighbourhood trust	p_β (1-unit increase)	$+0.024$	0.029	.410
Neighbourhood trust	p_γ (1-unit increase)	$+0.119$	0.036	$.001^{***}$

Notes. RE probit (GHQ-12, nAGQ = 12). Posterior probabilities as continuous predictors. Controls: age, sex, swb, employment, couple status, institutional trust, wave dummies. $N = 3{,}891$. $\dagger p < .10$; $*p < .05$; $**p < .01$; $***p < .001$.

S6 Sensitivity: Active-Only Membership Coding

The main analysis codes membership as active *or* inactive (`c12` ≥ 2). To assess sensitivity to this coding choice, models should be re-estimated using strict active-only membership (`c12` $= 3$). This requires re-running `02_latent_markov.R` with `MEMBER_CODE_LOGIC = "active_only"` and is reserved for the revision stage.

Planned analyses:

1. Latent Markov Model fit and profile comparison (active-only vs. any-member)
2. Portfolio-position assignment comparison (cross-tabulation)
3. Trust AMEs under active-only classification
4. Exit-from-γ SES gradient under active-only coding

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